Capstone Project - Financial News Sentiment Analysis

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Introduction

need a proper introduction to the topic and THEN relating it to a data science problem

Predicting the stock market is a topic of great intrest to many people. News and consumer sentiment play a crucial role in influencing stock prices, as traders continuously monitor financial news and social media to assess supply, demand, and market trends. This process requires constant tracking, interpretation, and decision-making to respond effectively to market fluctuations. It is therefore evident that an NLP system could be of great use, being able to (1) summarise articles and identify key information and (2) provide sentiment variables as input features for a stock price prediction model.

This report presents a pipeline that collects news articles using web scraping techniques and processes them through NLP systems to extract meaningful insights. All code and reporting files are available on GitHub.

Web Crawler

Data Type Selection

It is clear that a textual data corpus is required for the exploration of NLP tecquniques in this setting. There are numerous sources on the world wide web that were considered.

The first source that was considered was the ASX market announcements [5]. The announcements are straight from the source, come at a high frequency, but come with the drawback of sometimes being mundaine and not helpful, the format being in a pdf document which is not conducide to text mining with tool such as `BeautifulSoup`, and it being against the terms of use of the ASX [6].

The second source of data that was considered was social media platforms, in particular Twitter and Reddit. Using social media platforms would afford a large quantity of data, but with each document itself being small. Using social media also has the drawback of being unstructured, and also the text data using colloquial language and having spelling mistakes, with extensive preprocessing required as found in many papers put the papers here, say something about how social media users arent held to a standard to be relied upon for factual or accurate information

In contrast, financial news articles are more factual than socual media, with the added benefit of being longer in length, having correct spelling and grammar. Furthermore, websites can offer more structure in the way they present their articles, with easy ways to scrape key metadata, perhaps reference the next section where this is elaborated on.

Website Selection

Table 1 shows the comparison of various websites that contain financial and market-related articles, evaluating their suitability for scraping.

The smallcaps website was selected for the following reasons: - no dynmaic content, so no elaborate web scraping web driver is required, cna just use `BeautifulSoup` - the website is all about stocks, so no need to filter out non-stock related articles - well labelled articles with stock codes

there is however drawbacks to this particular website and especially only choosing one website from which to scrape from.

asdf

Website Name (URL hyperlinked)	**Scraping Permitted in Terms of Use?**	**Allows Article Scraping in Robots.txt?**	**Uses Cloudflare?**	**Notes of quality**
Market Index	No	Yes	No	Company tags on articles
Yahoo Finance	No	No	No	no mentioned stock labelling. General politics also included which is difficult to relate to individual stocks
Trading Economics	Yes	Yes	No	General politics and economics also included — would require filtering (unstructured)
Financial Review	No	No	No	
The Motley Fool	No	Yes	No	
Sydney Morning Herald	No	No	No	
The Market Online	No	Yes	No	
ABC News	Yes	Yes	No	Regular news mixed with financial news, difficult to use, untagged stock market
Small Caps	Yes	Yes	No	Sitemap highly conducive to scraping. Stock codes provided in articles. All about stocks — no general politics. Static web content (use BeautifulSoup)
Morning Star	No	Yes	No	No sitemap provided
Australian Stock Report	Yes	Yes	No	Unstructured in comparison to other article sites
Stock Head	Yes	Yes	No	Great sitemap but no list of stock codes in article
The Australian	No	No	No	
Hot Copper	No	Yes	No	

Table 1: Comparison of financial and market-related websites for article scraping suitability. Green rows indicate favourable conditions for scraping.

Web Scraping

To begin this analysis, we scrape the sitemap of `smallcaps.com.au`. This offers the benefit of having a consolidated list of all article URLs, eliminating the need to emulate human navigation through pages of website search results using `Selenium`. The `lxml` parser is used for this task due to its fast performance and capability to handle both XML and HTML scraping. table 2 shows the distribution of the last modified date of the urls in the XML sitemap.

Article Year	**Number of Observations**
2023	10,468
2024	2,374
2025	323
Total	**13,165**

Table 2: Number of Observations per Year

Now, individual article content needs to be scraped. The following listing uses `BeautifulSoup` and the `requests` library to extract the contents of a given URL, which was sourced from the sitemap. The `lxml` parser was chosen again for its speed, and since the necessary external C dependency had already been installed with the `lxml-xml` parser during the XML sitemap scraping, no additional setup was required [7]

The contents of the HTML were analysed and parsed into a dictionary capturing the title, author, date, sector, stock codes, and the article body, as demonstrated in the listing below. `BeautifulSoup` was used

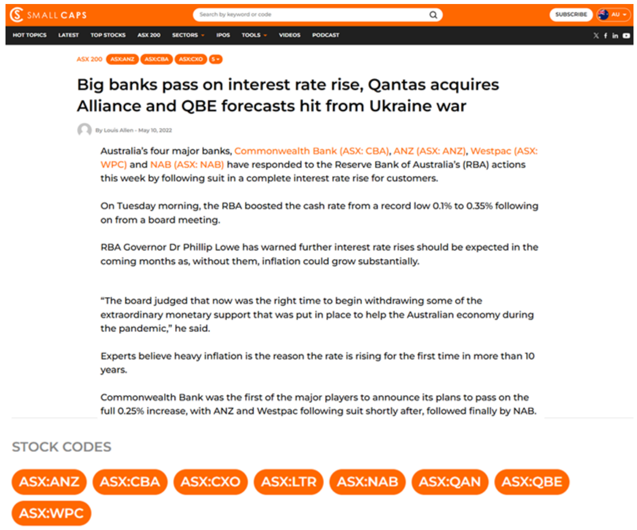

Fig. 1. Example Article [1], containing fields such as title, author, date, and stock codes

to extract data from the relevant tags. Below is an example output from the parsing process:

Listing 1: Example HTML extraction code

```
soup_test = extract_html_from_url('https://smallcaps.com.au/big-banks-interest-rate-rise-qantas-acquires-alliance-qbe-forecasts-hit/')
parsed_dict = parse_html_to_dict(soup_test)
parsed_dict
```

Key	Value
'title'	'Big banks pass on int...'
'author'	'Louis Allen'
'date'	'May 10, 2022'
'sector'	'ASX200'
'stock_codes'	['ASX: CBA', 'ASX: ANZ', ...]
'document'	"Australias four maj..."

Table 3: Parsed dictionary converted into a table

Exploratory Data Analysis

Table 3 depicts the structure of the data. Collected from smallcaps was 1532 articles. Figure 2 depicts the distribution of the scraped articles. The reports sector label is visualised in fig. 4, provided by the smallcaps website. The industry label is provided by the ASX website [2]. It is evident the high proportion of articles in the mining sector, potentially leading to bias in any investigation that is conducted in this data without any other intervention.

Analysing the textual corpus, fig. 3 shows the distribution of document length, where it can be seen that the mean document wordcount is 409 words, which combined with the quantity of data collected should be sufficient for training a machine learning model a source for this would be better than hopes and prayers

Feature Extraction

Sentiment Analysis

We model the sentiment of an article using the VADER (Valence Aware Dictionary and sEntiment Reasoner), as this has shown to having promising results in a similar setting, as found by [8]. This implementation of vader is assisted through a Geeks for Geeks instructional article [9]

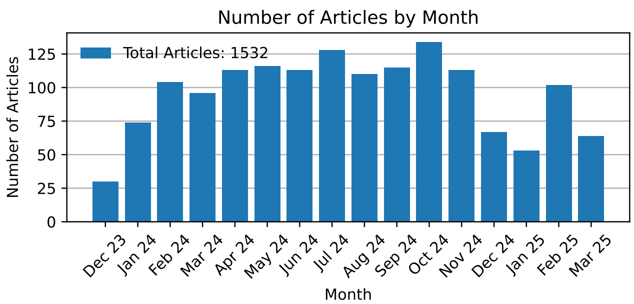

Fig. 2. Quantity of Scraped Articles per Month

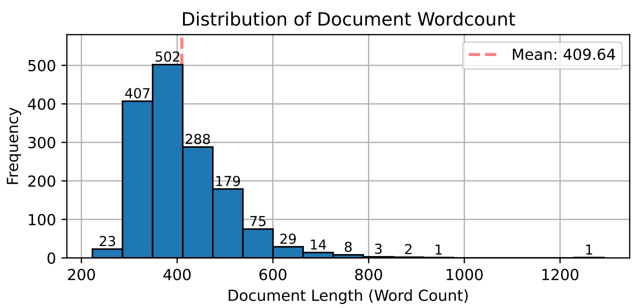

Fig. 3. Distribution of Document Length

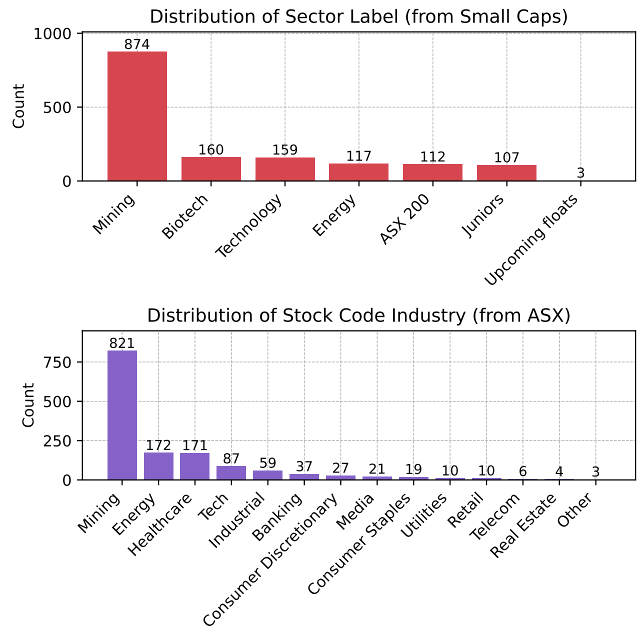

Fig. 4. Distribution of Articles by Sector, using the *smallcaps* industry label and the ASX sector label [2]

VADER's rule-based approach relies on fixed lexicons and doesn't fully capture the context-dependent sentiment embedded in text, whereas BERT's deep contextual embeddings allow it to discern and represent nuanced sentiment more effectively. FinBERT is a domain-specific adaptation of the BERT model, fine-tuned on financial texts to accurately determine the sentiment expressed in this particular context [10]. This model was retrieved using the huggingface library [11]

While this sounds better, it relies on the data being labelled, making training on a large corupus of data unfeasible. We therefore utilise the finbert-tone pretrained transformer which is the FinBERT model trained on 10,000 manually annotated documents (positive, negative, neutral) sentences from analyst reports.

Machine Learning

After having sraped the articles for the documents and sentiments? a machine learning model was developed to predict stock price. figs. 5 and 6 show the structure of a typical feed forward neural network and a recurrent neural network. Recurrent neural networks (RNNS), and others like RNNS, are particularly well suited to sie seues data

[8] show that an LSTM model, when trained on timeseries price data and sentiment data from news articles, can outperform a model trained on price data alone. Analysis of the literature suggests that SVM models, whilst the most popular [12] often underperformed in a financial setting, due to parameter optimisation being required [12, 13]

Talk about the adam optimiser According to Kingma et al., 2014, the method is "computationally efficient, has little memory requirement, invariant to diagonal rescaling of gradients, and is well suited for problems that are large in terms of data/parameters". [14, 15]

the events extracted from the news are usually quite sparse, resulting in unreliable and unsustainable predictive power [13] (bar for bar)

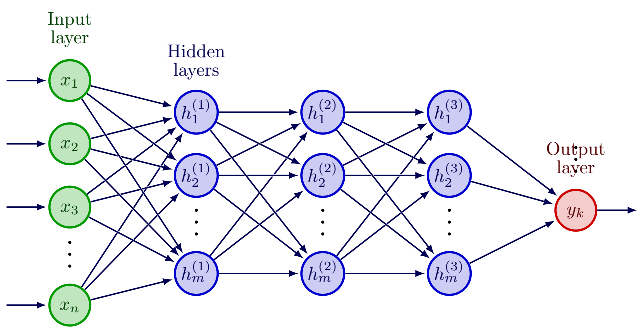

Fig. 5. Typical "FeedForward" Neural Network Topology [3]

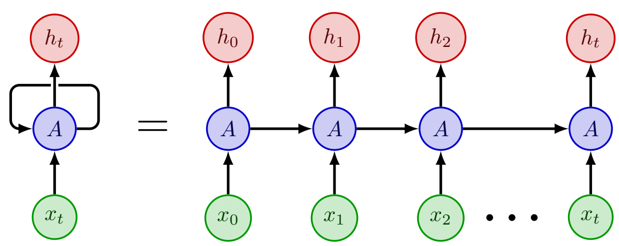

Fig. 6. Recurrent Neural Network Topology [4]

References

[1] L. Allen. (2022, May) Big banks pass on interest rate rise, qantas acquires alliance and qbe forecasts hit from ukraine war. https://smallcaps.com.au/big-banks-interest-rate-rise-qantas-acquires-alliance-qbe-forecasts-hit/. Accessed: 2025-04-02.

[2] ASX Limited, "Company directory," 2024, accessed: 2024-04-08. [Online]. Available: https://www.asx.com.au/markets/trade-our-cash-market/directory

[3] IWN, "Drawing neural network with tikz," https://tex.stackexchange.com/questions/153957/drawing-neural-network-with-tikz, October 2021, answer posted on TeX StackExchange, Accessed: 2025-04-07. [Online]. Available: https://tex.stackexchange.com/questions/153957/drawing-neural-network-with-tikz

[4] user121799, "How do i draw a simple recurrent neural network with goodfellow's style?" https://tex.stackexchange.com/questions/494139/how-do-i-draw-a-simple-recurrent-neural-network-with-goodfellows-style, June 2019, answer posted on TeX StackExchange, Accessed: 2025-04-07. [Online]. Available: https://tex.stackexchange.com/questions/494139/how-do-i-draw-a-simple-recurrent-neural-network-with-goodfellows-style

[5] ASX Limited, "Today's announcements," https://www.asx.com.au/markets/trade-our-cash-market/todays-announcements, 2025, accessed: 2025-04-02.

[6] ——, "Terms of use," https://www.asx.com.au/legals/terms-of-use, 2023, last updated 12 October 2023. Accessed: 2025-04-02.

[7] L. Richardson, *Beautiful Soup Documentation*, 2004–2025, retrieved April 27, 2025. [Online]. Available: https://www.crummy.com/software/BeautifulSoup/bs4/doc/#installing-a-parser

[8] C.-R. Ko and H.-T. Chang, "Lstm-based sentiment analysis for stock price forecast," *PeerJ Computer Science*, vol. 7, no. 1, p. e408, March 2021.

[9] GeeksforGeeks, "Python sentiment analysis using VADER – using python," December 2024, accessed: 2025-04-08. [Online]. Available: https://www.geeksforgeeks.org/python-sentiment-analysis-using-vader/

[10] D. T. Araci, "Finbert: Financial sentiment analysis with pre-trained language models," Master's thesis, University of Amsterdam, 2019, preprint available on arXiv. [Online]. Available: https://doi.org/10.48550/arXiv.1908.10063

[11] ProsusAI, "Finbert [model]," n.d., retrieved March 27, 2025, from Hugging Face. [Online]. Available: https://huggingface.co/ProsusAI/finbert

[12] A. Y. Al-Dulaimi, S. N. Al-Saadi, and S. A. Jasim, "The stock exchange prediction using machine learning techniques: A comprehensive and systematic literature review," *Jurnal Ilmu Komputer dan Informasi*, vol. 14, no. 2, pp. 132–150, July 2021, accessed: 2025-04-07. [Online]. Available: https://www.researchgate.net/publication/353002279

[13] X. Zhang, Y. Li, S. Wang, B. Fang, and P. S. Yu, "Enhancing stock market prediction with extended coupled hidden markov model over multi-sourced data," *arXiv preprint arXiv:1809.00306*, September 2018, 19 pages. Submitted on 2 Sep 2018. Accessed: 2025-04-07. [Online]. Available: https://doi.org/10.48550/arXiv.1809.00306

[14] Keras Team, "Adam class api docs," https://keras.io/api/optimizers/adam/, accessed: 2025-04-07.

[15] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2015, published as a conference paper at the 3rd International Conference for Learning Representations, San Diego, 2015. [Online]. Available: https://doi.org/10.48550/arXiv.1412.6980